



GeoPops: An open-source package for generating geographically realistic synthetic populations

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Atlantic Coast Center for Infectious
Disease Dynamics and Analytics



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Epidemic Scenario Modeling



- We know there are **demographic and geographic subgroup differences** in human behavior and health and economic outcomes during infectious disease epidemics.
- **How can we capture this in models?**
 - *Agent-based modeling on synthetic populations is one way*
- **Why is this important?**
 - *Inform targeted interventions and efficient resource allocation*
 - *If infections start in this county, where will they spread next? Who will be most affected?*
 - *Who should we vaccinate first? What happens if we don't vaccinate?*
 - *Should we focus on prevention or treatment/care?*

GeoPops



- **Open-source package** for generating geographically and demographically realistic **synthetic populations** for any US Census location using publicly available data
- **Goal: User-friendly tool for context-specific scenario modeling** of respiratory diseases
- Same methodology as [GREASYPOP-CO](#) (One Health Trust) but wrapped in convenient Python package that can be pip installed
- Automatic data downloading
- User only uses Python
- Compatibility with recent Census data (developing)
- Compatibility with ABM software [Starsim](#) (IDM)
for simulating infectious and non-communicable diseases

github.com/ACCIDDA/GeoPops

```
pip install geopops
```

WSC Presentation



1. GeoPops Methodology
2. Make a population of Maryland, USA
3. Run SIR model of generic upper respiratory infection with Starsim
4. Track outcomes by age group and county
5. Repeat with Geo-Synthetic-Pop (Jiang et al. 2024)

github.com/ACCIDDA/GeoPops/tutorials/WSC

Larger Project



- SARS-CoV-2 model with SEIRD structure and age-dependent transmissibility and susceptibility
- Simulate first wave of the pandemic in Maryland
- Using populations made with GeoPops and two other generators
- Compare agents and households to Census data
- Compare network statistics of each population
- Compare simulated disease outcomes to observed data

GeoPops: Agents and networks



Agent attributes

- Age
- Gender
- Race/ethnicity
- Worker status
- Student status
- School grade
- Commuter status
- Income category
- Workplace category
- Home CBG
- School CBG if applicable
- Work CBG if applicable
- GQ CBG if applicable



School Network

- List of students for each
- Size of grade levels
- Number of teachers
- CBG locations



Household Network

- List of agents for each
- CBG locations



Workplace Network

- Number of workers
- CBG if inside population area



Group quarters (GQ) Network

- Number of residents
- Number of staff
- CBG locations

- Common network structure when modeling respiratory infections
- Few open-source methods for making this and fewer that are user-friendly

Step 1: Generating agents and households



Public Use Microdata Samples (PUMS)



- Subset of census survey responses
- Compositional details, e.g., the number of 5-member households in a CBG with school-aged children

Combinatorial
optimization
(CO)

American Community Survey (ACS)



- Demographic data at fine geographic scale
- Limited to marginal counts, e.g., the number of households in a CBG with 5 members and the number of households in a CBG with school-aged children

Alexander Y. Tulchinsky, Fardad Haghpahan, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. <https://arxiv.org/abs/2406.14698>

Step 2: Assigning agents to schools and workplaces



	Data source	Data	Methods
Schools	National Center for Education Statistics (NCES)	Student and staff counts for each grade level in public schools	• Students assigned to closest school that offered required grade • Teachers assigned to school by selecting teachers from nearby CBGs
Workplaces	Longitudinal Employer-Household Dynamics (LEHD)	Origin-destination and workplace characteristics	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers

Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. <https://arxiv.org/abs/2406.14698>

Step 3: Generating Contacts



Layer	Algorithm	Parameters
Household	All agents within a household are fully connected	N/A
School	Stochastic Block Model within each school	- Assortativity: $\alpha=0.9$ - Mean degree: $K=12$ - Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	- Assortativity: $\alpha=0.9$ - Mean degree: $K=8$ - Blocks are income levels ($<40k$ or $\geq 40k$)
Group quarters	Watts-Strogatz Model within each GQ facility	- Mean degree: $K=12$ - Rewiring probability: $\beta=0.25$

Alexander Y. Tulchinsky, Fardad Haghighpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. <https://arxiv.org/abs/2406.14698>

Code example: MD, DC, northern VA



```
pars_geopops = {'path': "YOUR_OUTPUT_PATH",          # Set a folder where you want output files to be stored
                'census_api_key': "YOUR_CENSUS_API_KEY", # Your Census API key
                'julia_env_path': "YOUR_JULIA_ENV_PATH", # Path to your Julia environment
                'main_year': 2019,                      # Year of data
                'geos': ["24",                          # State or county fips codes making up your population area
                        "11",
                        "51013", "51059", "51061", "51107", "51153", "51179", "51510", "51600", "51610", "51630", "51683", "51685"],
                'commute_states': ["24", "11", "51", "10", "42", "54"], # State fips of commute data to download
                'use_pums': ["24", "11", "51", "10", "42", "54", "25", "34", "37", "09"]} # State fips of PUMS data to download

geopops.WriteConfig(**pars_geopops) # Define parameters for pop generation in config.json
geopops.DownloadData()              # Download raw data files
geopops.ProcessData()               # Preprocess data
j = geopops.RunJulia()               # Run Julia portion
j.run_all()                         # Run all methods (CO(), SynthPop(), Export()) in RunJulia() class
```

N ~9M. Several hours on my laptop, but you only have to run it once (a year)!

Accuracy

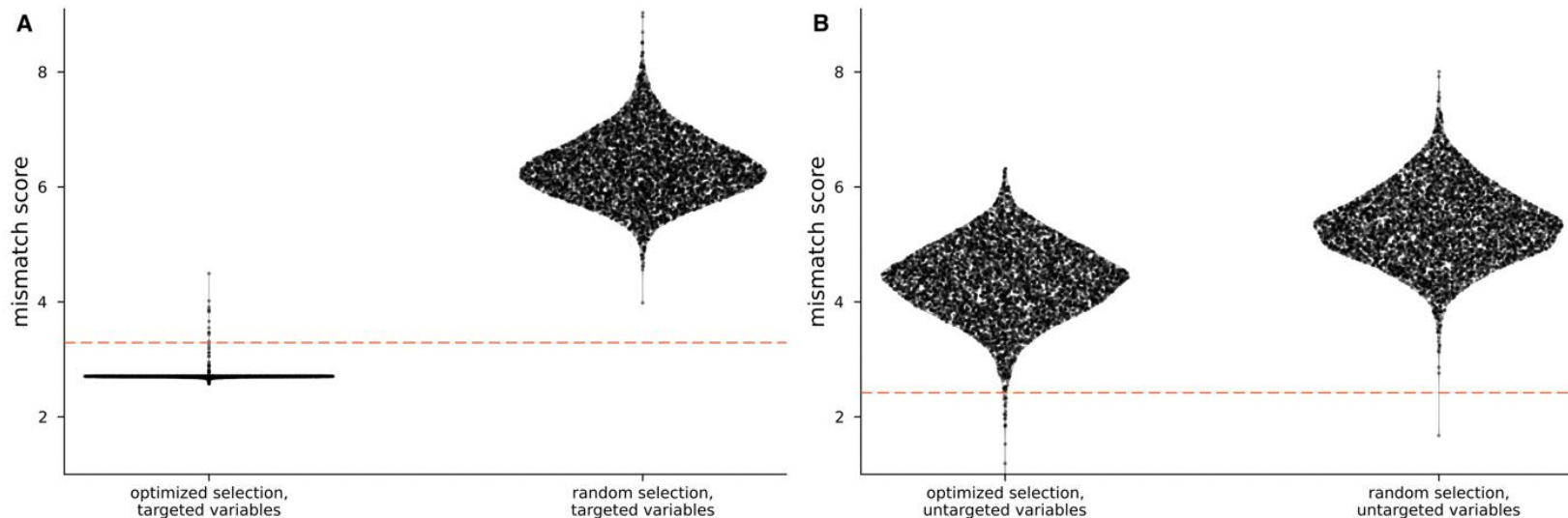
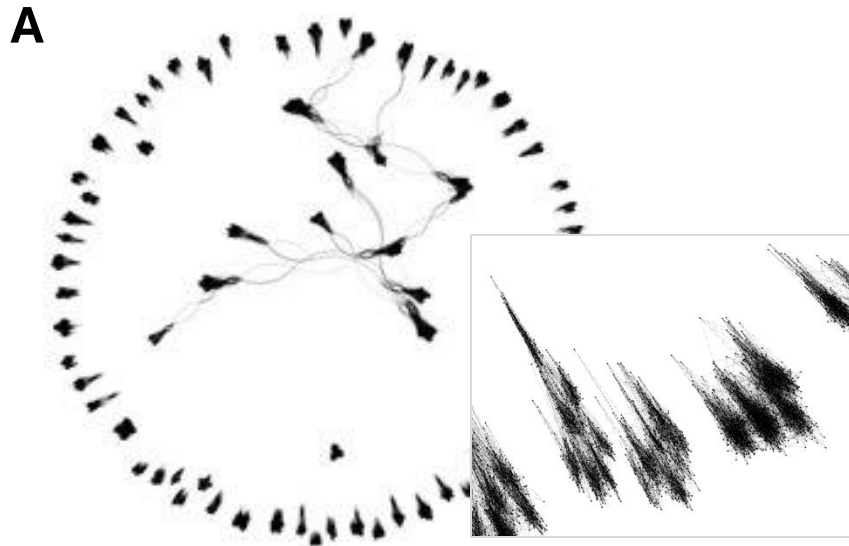


Fig 1. Fit between census data and households selected by combinatorial optimization for each census block group (CBG) in the Maryland-area synthetic population. (A) Compared to a random selection of households (right), the optimized selection (left) is a much better match to the Census for the variables used to guide the CO algorithm (the targeted census variables). (B) The optimized selection (left) is still a better match than the random selection (right) for additional census variables that were not targeted.

Tulchinsky et al. 2024

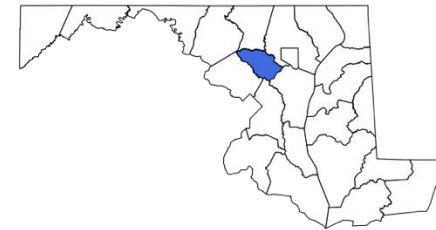
School Network Visualization (Howard County, MD)



B

p1	p2	beta
748	78	1
984	883	1
1702	115	1
1709	1521	1
1875	651	1

Visualization and table snippet of the school network. (A) Nodes are students and teachers clustered in grades within schools. (B) Same graph in Starsim tabular format where beta is the weight of the connection between person one and person two.



GeoPops with Starsim



Modules passed into Starsim to simulate disease transmission

**GeoPops agents
(ss.People)**

SIR model (ss.Disease)

- Respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

**GeoPops networks
(ss.Network)**

Subgroup Tracking (ss.Analyzer)

- Infections by age and county

**Simple example with
generic upper
respiratory infection**

Code Example



```
# Define epidemic parameters
pars_ss = sc.objdict(start = 0, stop=50, # Simulation duration
                    rand_seed = 123, # random seed
                    diseases = dict[str, str | float](type = 'sir', # type of disease
                                                    init_prev = 0.01, # initial prevalence
                                                    beta = 0.1,      # transmission rate
                                                    dur_inf = 7,      # duration of infection
                                                    ))
```

```
# Run SIR model with GeoPops people, networks, and subgroup tracking
sim_gp = ss.Sim(pars_ss, # Dictionary of SIR disease structure and parameters
               people=ppl_gp, # People object created with geopops.ForStarsim.People()
               networks=[h_gp, s_gp, w_gp, g_gp], # Created with geopops.ForStarsim.GPNetworks()
               analyzers=[age_tracking, county_tracking], # Created with geopops.ForStarsim.SubgroupTracking()
               ).run()
res_gp = sim_gp.results
```

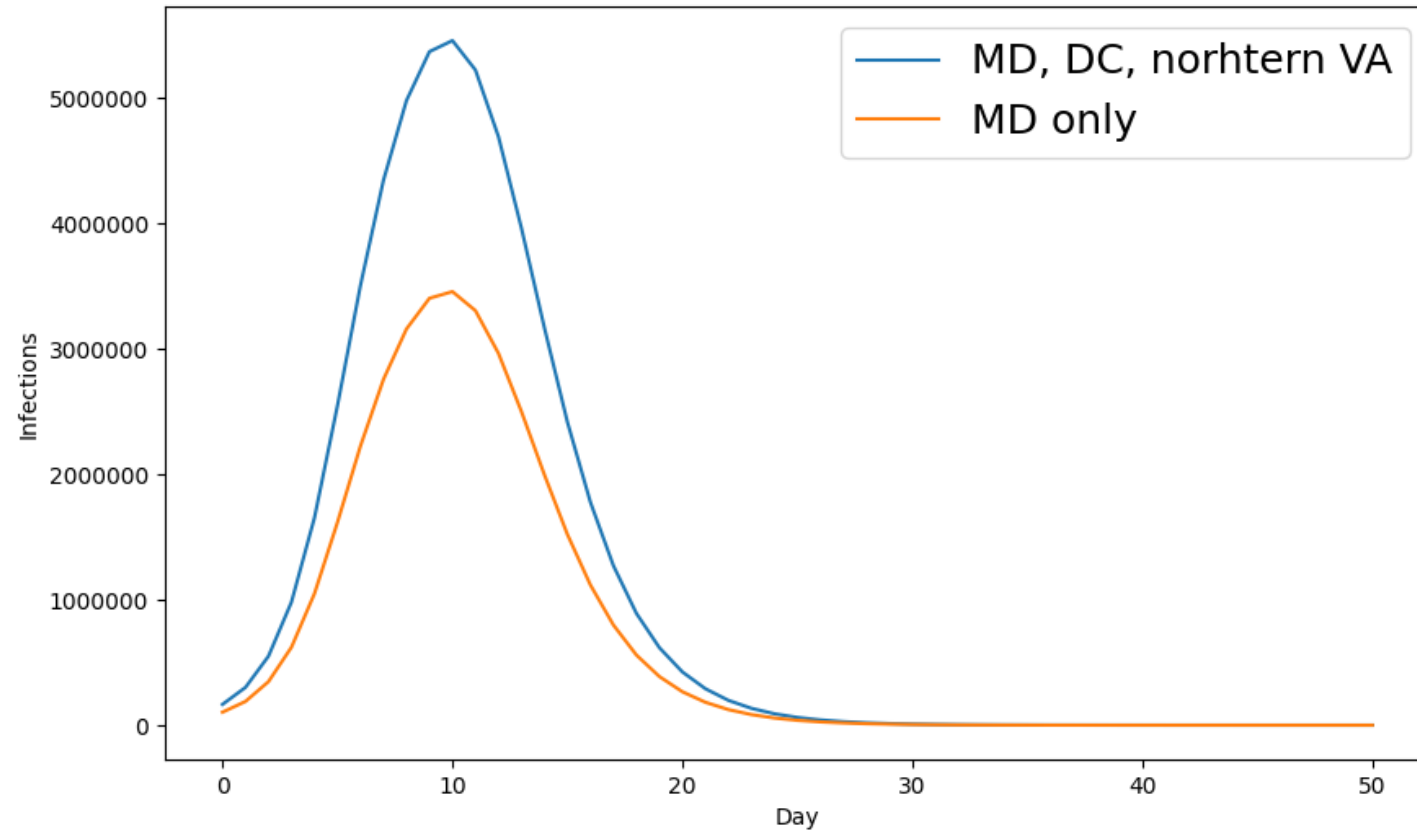
✓ 12m 12.8s

N ~9M

Infections



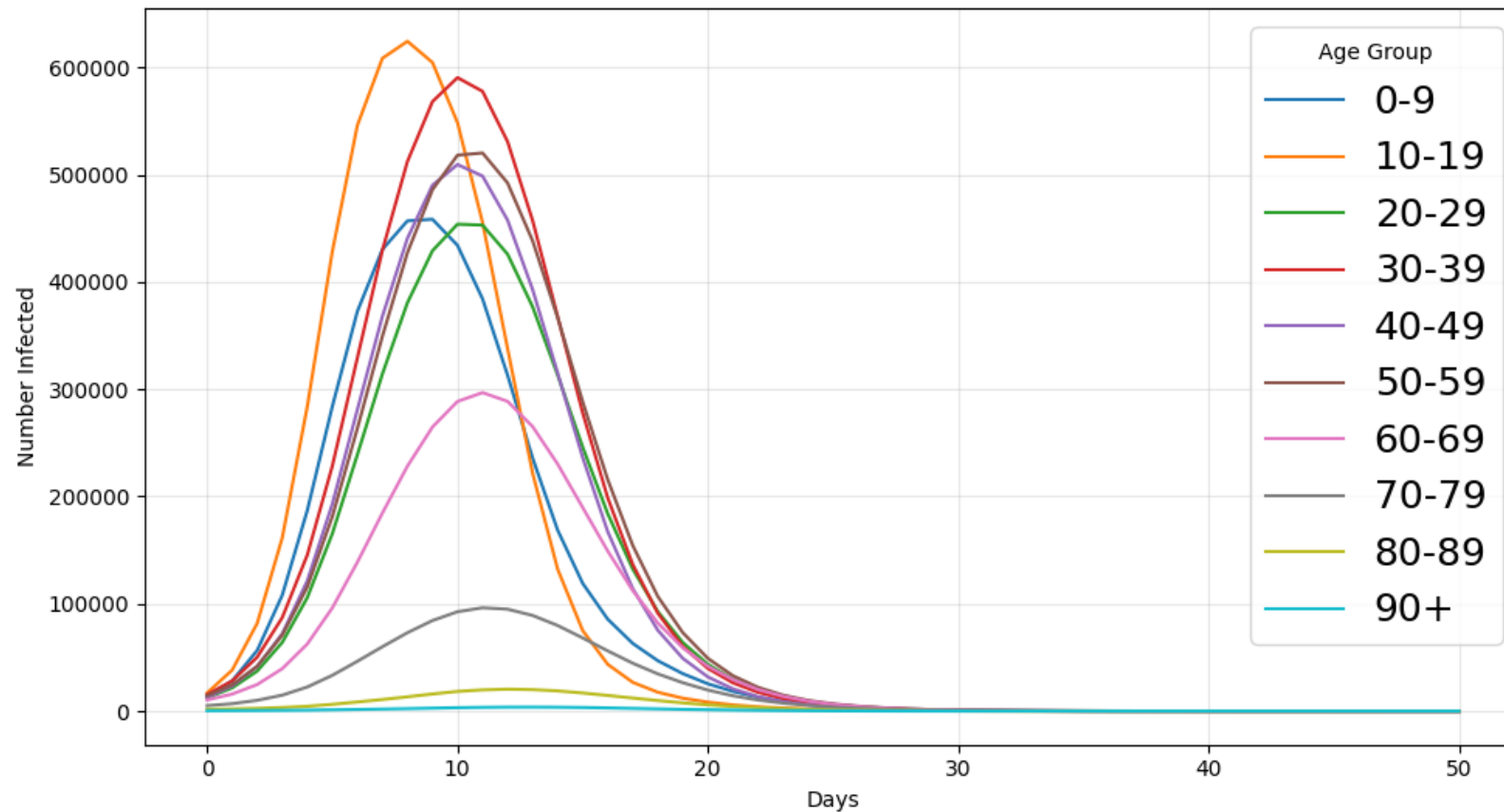
Infections over time



Outcome tracking by demographic subgroup



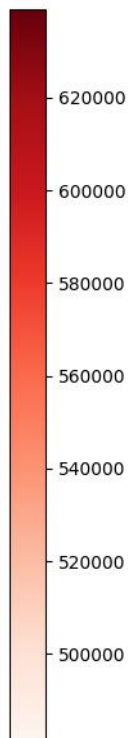
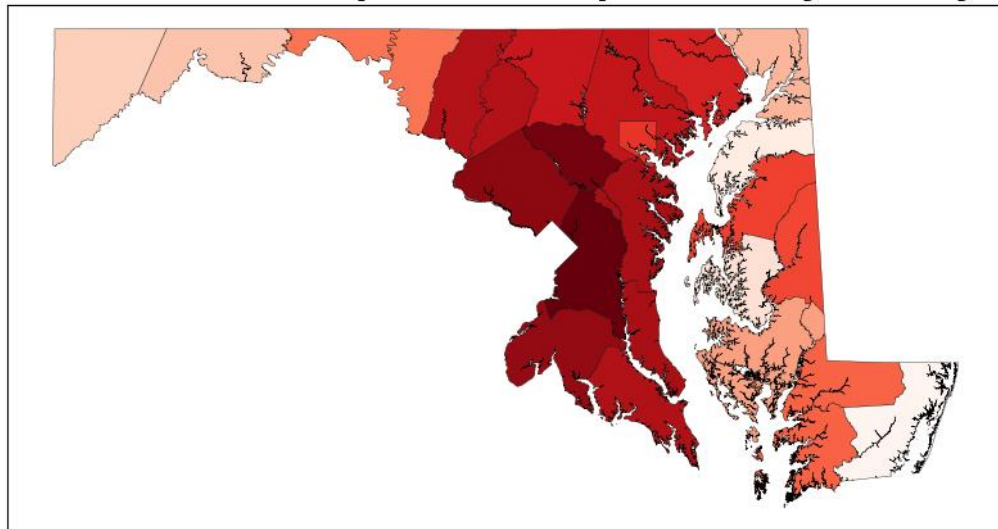
Infections by Age Group, MD



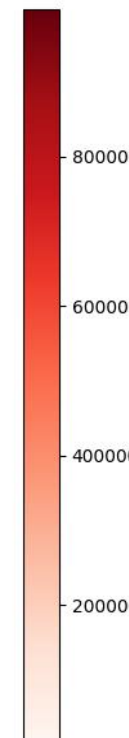
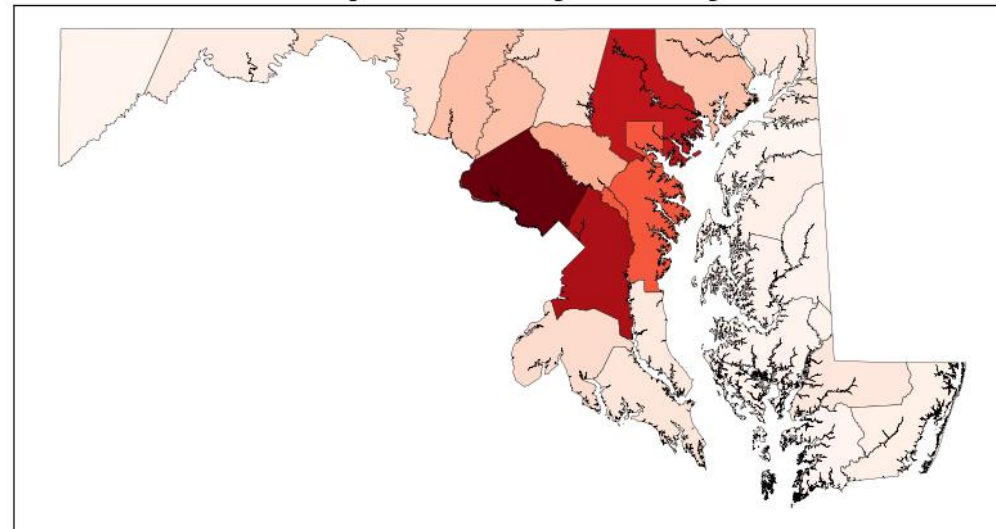
Outcome tracking by geographic subgroup



Total Infections per 100k Population by County



Population by County



Why is this cool?!



Open-source, user-friendly framework for demographic and geographic subgroup tracking

- Don't have to build a model from scratch
- Can be used by state and county health departments for context-specific scenario modeling
- Seed infections to a specific geography or target interventions to specific demographic or geographic groups
- **Inform decisions about targeted interventions, which can lead to more efficient resource use and reductions in overall disease burden and disparities**
- There are several research groups already using Starsim to model respiratory diseases that can now use GeoPops as well

Actively maintained



Comparison to Synthetic Populations



Looking for...

- Open-source (downloadable datasets or reproducible code)
- Multiple demographic attributes
- CBG granularity (at least county)
- Home, school, workplace networks as graphs

Close, but limitations (for my purposes)...

- Geo-Synthetic-Pop (University at Buffalo)
- UrbanPop (Oak Ridge)
- SynthPops (Institute for Disease Modeling)
- FRED (RTI International)
- Others
 - Not published or open-source
 - Only for specific city/area
 - I'm not aware of (email me!)



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A Large-Scale Geographically Explicit Synthetic Population with Social Networks for the United States

[Na Jiang](#) ✉, [Fuzhen Yin](#), [Boyu Wang](#) & [Andrew T. Crooks](#)

<https://doi.org/10.1038/s41597-024-03970-1>

Pros

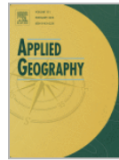
- Pre-built state populations
- All python
- Completely open-source

Cons

- Doesn't work beyond 2019, Census boundaries changed
- No longer maintained (last activity over a year ago)
- State populations do not include out-of-state commuters in workplaces
- Fewer demographic attributes (only age, sex, and urbanicity)



Applied Geography
Volume 151, February 2023, 102844



UrbanPop: A spatial microsimulation framework for exploring demographic influences on human dynamics

Joseph Tuccillo^a  , Robert Stewart^a, Amy Rose^a, Nathan Trombley^b, Jessica Moehl^a,
Nicholas Nagle^c, Budhendra Bhaduri^a

<https://doi.org/10.1016/j.apgeog.2022.102844>

Pros

- All in python, actively maintained, responsive
- Really detailed re demographics
- Private and public schools with school names

Cons

- Need to be pretty good python programmer
- Code for assigning agents to schools and workplaces is not yet public
- Resulting dataset needs extra processing to be comparable to GeoPops
 - Agents within schools and workplaces are not connected with graph algorithms
 - Group quarters residents have household size of 1 and are not mapped to specific facilities
 - Workers are not mapped to specific workplaces within a CBG
 - No minimum constraints on school size

SynthPops



<https://github.com/synthpops/synthpops>

Pros

- Pre-built populations for all states
- All in python, open-source

Cons

- Doesn't work with data beyond 2019
- No longer maintained
- Schools and workplaces do not have a geographical locations within the population area
- Fewer demographic attributes (only age and sex)
- Closed population (no out-of-state commuters)



Grefenstette et al. *BMC Public Health* 2013, **13**:940
<http://www.biomedcentral.com/1471-2458/13/940>



SOFTWARE

Open Access

FRED (A Framework for Reconstructing Epidemic Dynamics): an open-source software system for modeling infectious diseases and control strategies using census-based populations

John J Grefenstette^{1*}, Shawn T Brown², Roni Rosenfeld³, Jay DePasse², Nathan TB Stone², Phillip C Cooley⁴, William D Wheaton⁴, Alona Fyshe³, David D Galloway¹, Anuroop Sriram³, Hasan Guclu¹, Thomas Abraham³ and Donald S Burke¹

<http://www.biomedcentral.com/1471-2458/13/940>

Pros

- Pre-built populations

Cons

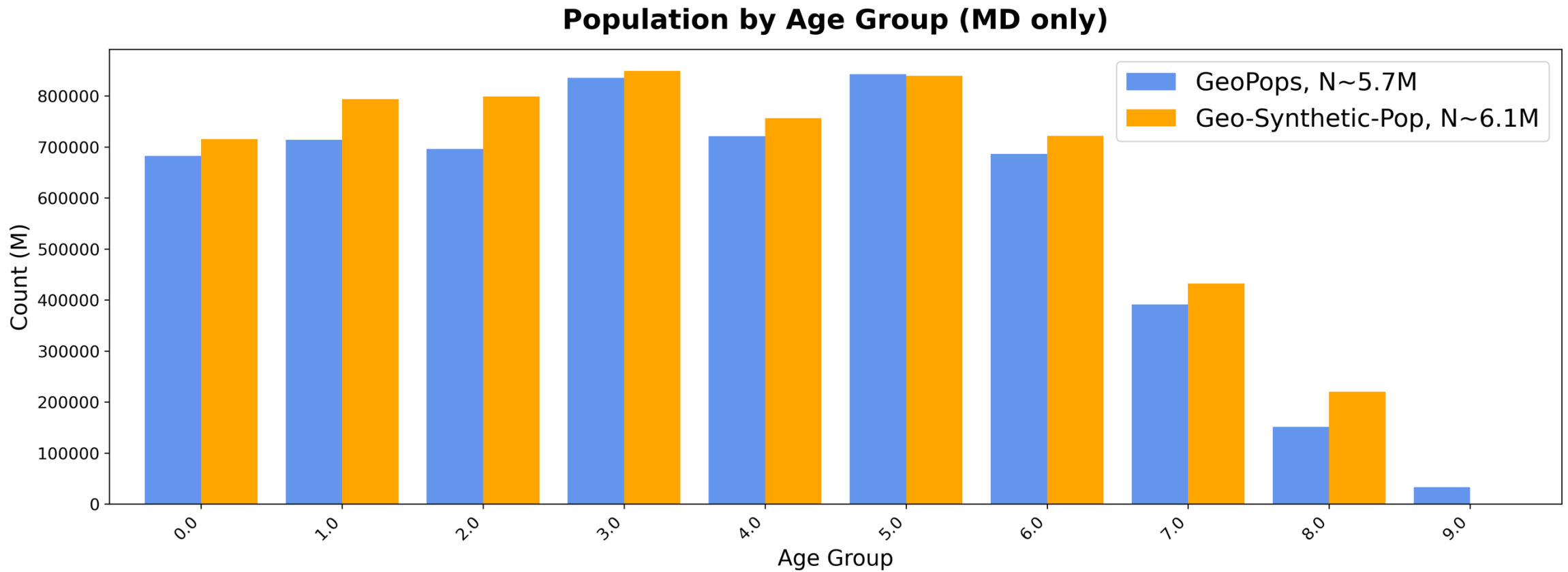
- C++
- Uses 2009 data
- No longer maintained

GeoPops vs Geo-Synthetic-Pop



	GeoPops	Geo-Synthetic-Pop
Agent and household generation	Combinatorial Optimization Age, sex, race/ethnicity, income, workplace category, CBG	Hierarchical Sampling Age, sex, urbanicity, Census Tract
School assignment	Assigned to closest school with capacity (NCES 2019)	Assigned to closest school with capacity (USEPA, Educational Institutions 2015)
Workplace assignment	LEHD origin-destination matrices and workplace characteristics (Iterative Proportional Fitting)	LEHD origin-destination matrices and workplace characteristics (Monte Carlo Sampling)

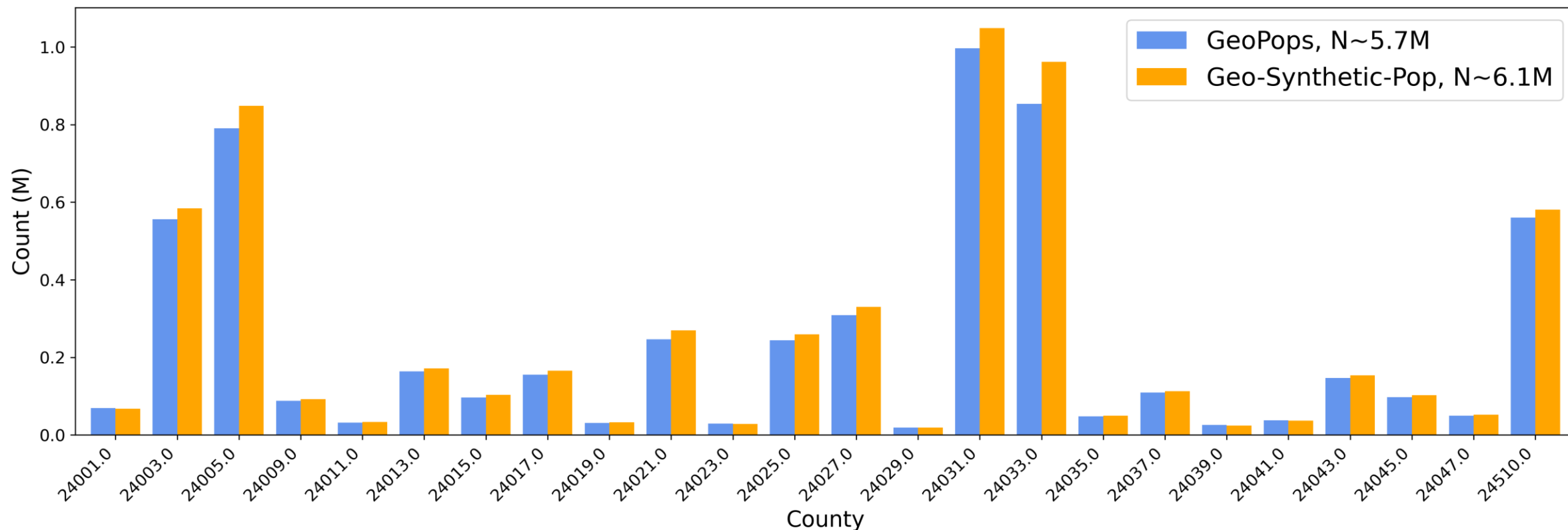
Population by Age Group (MD only)



Population by County (MD only)



Population by county (MD only)



GeoPops vs Geo-Synthetic-Pop

Network comparison



GeoPops = Blue (MD, DC, northern VA)

Geo-Synthetic-Pop = Orange (MD only)

Network	Algorithm	Mean degree (min, max)	No. Nodes	No. Edges
Home	Fully connected <5 Fully connected >5 Small-world	2.77 (1,19) 2.77 (1,11)	7,948,609 5,545,844	11,007,966 7,675,041
School	Stochastic Block Model <5 Fully connected >5 Small-world	12.01 (1,44) 5.20 (3,12)	1,762,561 1,014,435	10,583,502 2,636,664
Work	Stochastic Block Model <5 Fully connected >5 Small-world	7.55 (1,24) 3.53 (1,12)	4,557,404 1,814,870	17,193,880 3,201,197
Group Quarters	Small-world (Part of Home network)	12 (6,26)	200,873	1,205,292
Daycare	(Part of School network) <5 Fully connected >5 Small-world	5.20(1,13)	344,863	896,793

GeoPops vs Geo-Synthetic-Pop

Network comparison



GeoPops

- Lets you specify commuter states and creates placeholder agents for out-of-state workers that are nodes with edges in the work network
- Group quarters residents are assigned to specific group quarters facilities and then connected within facilities using the Small-world algorithm
- Can be multiple group quarters facilities per CBG, each represented as a Small-world graph

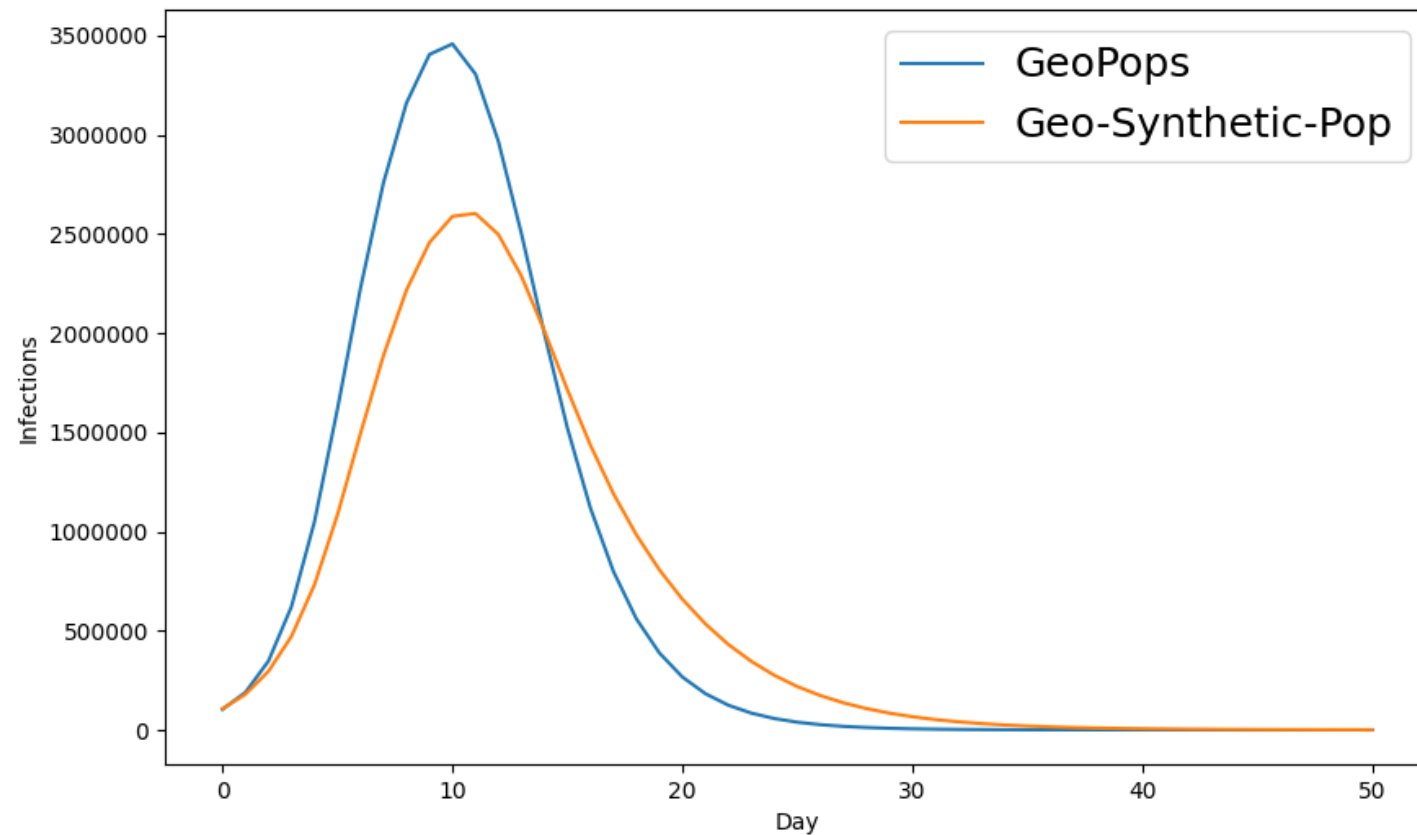
Geo-Synthetic-Pop

- Out-of-state commuters are not represented as nodes with edges in the work network
- Prisons, nursing homes, dorms, etc. are represented as one big “group-quarter household” per Tract, with the correct total population but no facility-level detail
- Only one Small-world graph per Tract

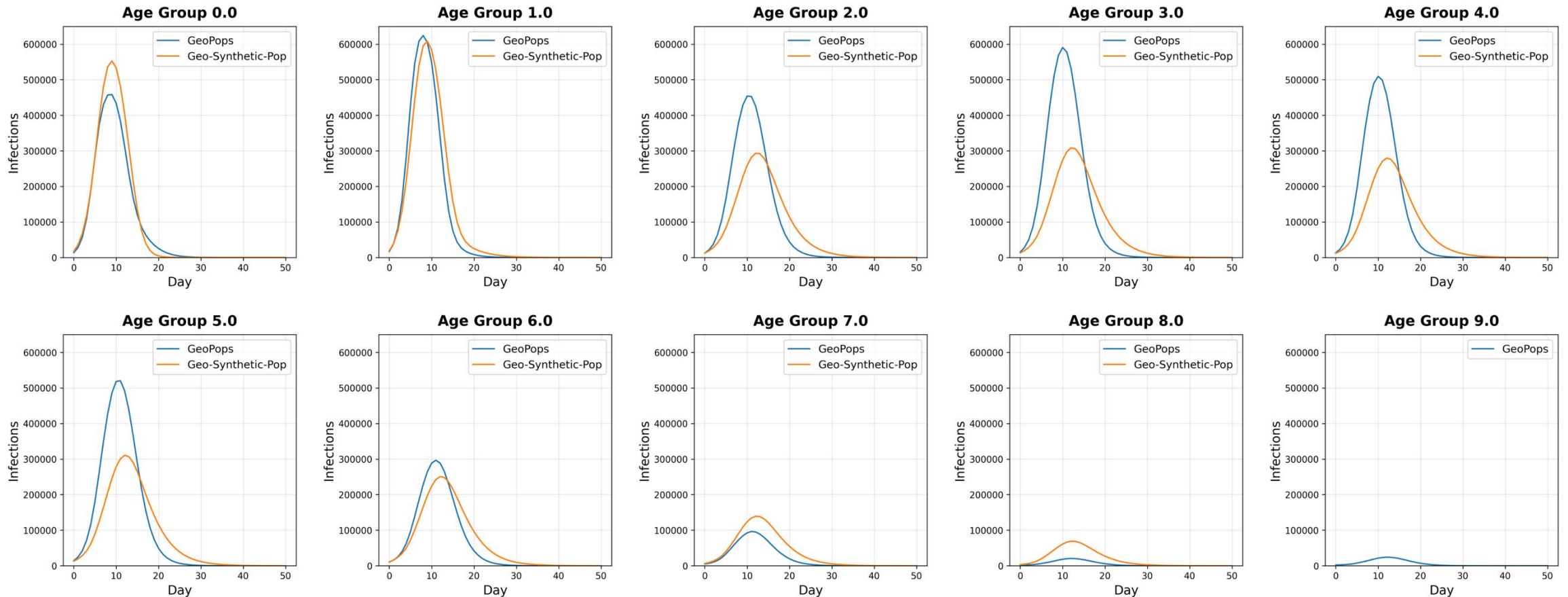
Comparison: Infections (MD only)



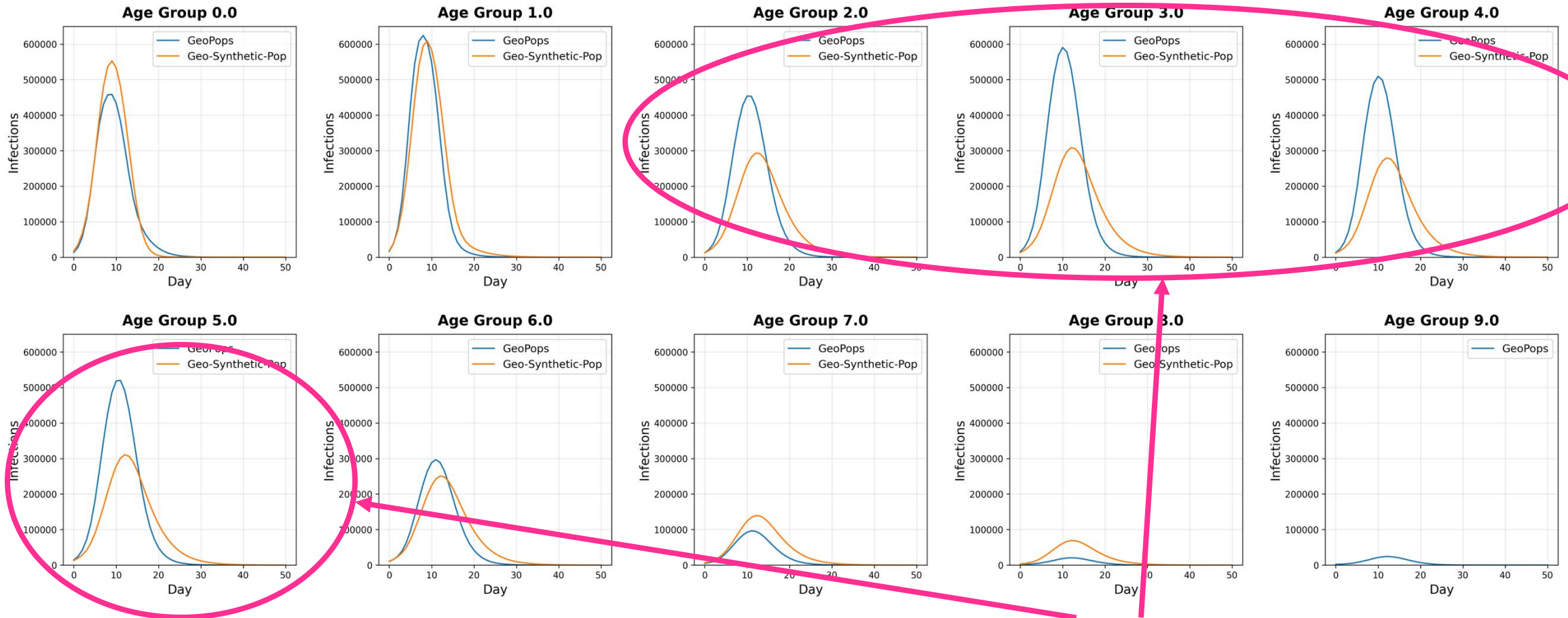
Infections over time



Comparison: Age Groups (MD only)

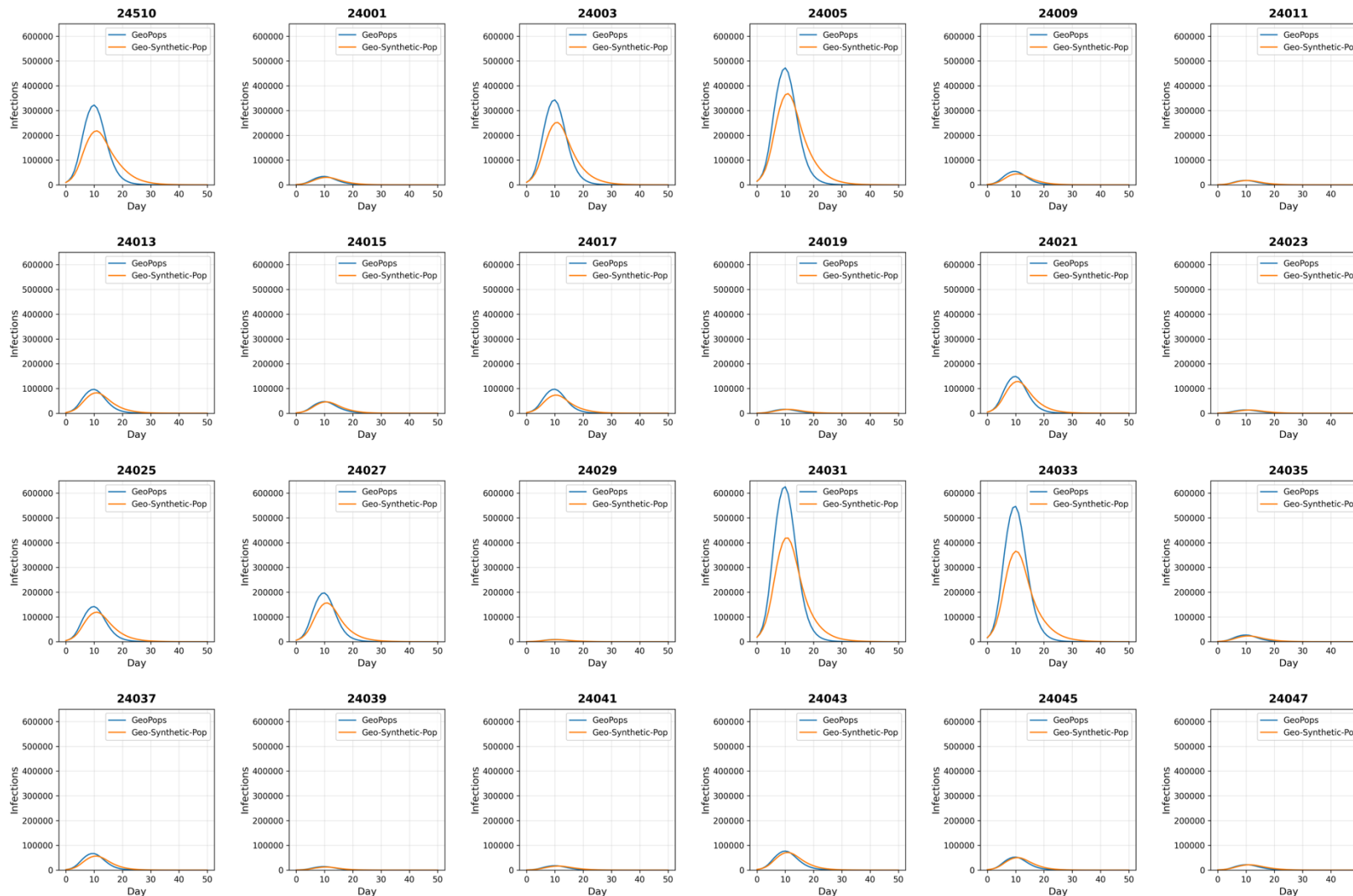


Comparison: Age Groups (MD only)



Working Age Groups

Comparison: Counties (MD only)



Comparison: Counties (MD only)



Montgomery and Prince George's counties are right next to DC



Larger Project



- SARS-CoV-2 model with SEIRD structure and age-dependent transmissibility and susceptibility
- Simulate first wave of the pandemic in Maryland
- Using populations made with GeoPops and two other generators
- Compare agents and households to Census data
- Compare network statistics of each population
- Compare simulated disease outcomes to observed data

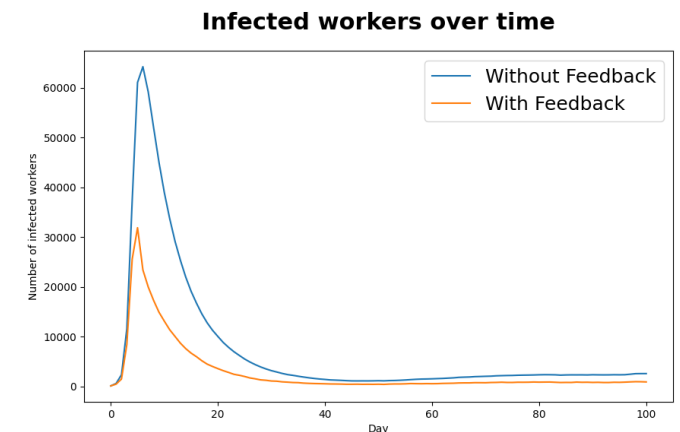
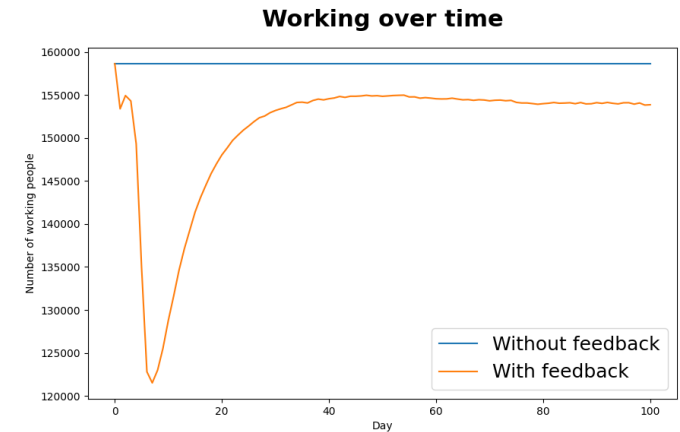
Applications: Endogenous Decision Making



Epidemics 2025

Modeling the impact of dynamic decision making on infectious disease outcomes by demographic and geographic subgroups: An open-source agent-based modeling framework

- Run Starsim COVID-19 model with **utility framework** on GeoPops population
- Individuals weigh **health-wealth trade-offs of staying home from work** as a function of their income and disease vulnerability
- Test different policy scenarios (e.g., paid sick leave, cash transfer) and compare to observed data from first wave of pandemic
- Demographic and geographic subgroup outcome tracking



Questions



- Pip install geopops and try going through the tutorial
- Log issues on GitHub! Please be patient 😊
- Poster, slides, and Notebook tutorial on GitHub

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github.com/ACCIDDA/GeoPops/tutorials/MIDAS

References



1. Hamilton A, Tulchinsky A, Lin G, Kerr C, Klein E, and Gardner L. 2025. GeoPops: Geographically Realistic Synthetic Population Generator. Johns Hopkins University and One Health Trust. <https://github.com/ACCIDDA/GeoPops>.
2. Kerr C, Stuart R, A Romesh, et al. 2024. “Starsim: A Fast, Flexible Toolbox for Agent-Based Modeling of Health and Disease.” <https://starsim.org/>.
3. Du H, Zahn MV, Loo SL, Alleman TW, Truelove S, et al. (2025) Improving policy design and epidemic response using integrated models of economic choice and disease dynamics with behavioral feedback. PLOS Computational Biology 21(10): e1013549. <https://doi.org/10.1371/journal.pcbi.1013549>



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